4 October 2026

Local-Environment Populations Govern Size-Dependent OER Activity in Supported Subnanometer IrO# Clusters

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Abstract

Proton-exchange-membrane water electrolysis is constrained by the high Ir demand of the acidic oxygen evolution reaction (OER), motivating supported subnanometer IrO# clusters that maximize Ir utilization. At this scale, however, changing cluster size simultaneously alters structure, oxygen coverage, and the distribution of nonequivalent active sites, making it difficult to identify the physical origin of size-dependent activity. Here, we combine systematic structure sampling, density functional theory, implicit solvation, and site-resolved microkinetics to resolve OER thermodynamics across Ir#O# (# = 1–30) clusters supported on anatase TiO2(101). The resulting activity landscape is strongly heterogeneous and nonmonotonic with cluster size, with substantial activity variations even among sites on the same cluster. This heterogeneity can be traced to the local oxygen coordination of the active Ir–#O motif. Increasing oxo-proximal coordination, quantified by CNO,co(Ir#), raises the #OH-to-#O free-energy cost while lowering that of #O-to-#OOH formation, thereby controlling the energetic balance between the two steps surrounding #O. Electronic-structure analysis associates this coordination dependence with charge redistribution at the binding Ir center, changes in Ir–O bonding, and reorganization of the Ir 5d–O 2p electronic structure. Steady-state microkinetics preserves the thermodynamic activity hierarchy, with ensemble rates governed primarily by the most active sites at each cluster size. Single-Ir dissolution thermodynamics further shows that higher local oxygen coordination is associated with greater resistance to Ir loss, extending the significance of local coordination from OER energetics to dissolution stability. These results establish local-environment populations as the structural origin of size-dependent catalytic behavior in subnanometer IrO# clusters, revealing local coordination rather than cluster size alone as the key structural variable underlying activity and stability.

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Keywords

acidic oxygen evolution reaction, iridium oxide, subnanometer clusters, local coordination, microkinetic modeling

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Local-Environment Populations Govern Size-Dependent OER Activity in Supported Subnanometer IrO_x Clusters

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[†]Guangsheng Liu and Leshen Zhang contributed equally to this work.
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Abstract. Proton-exchange-membrane water electrolysis is constrained by the high Ir demand of the acidic oxygen evolution reaction (OER), motivating supported subnanometer IrO_x clusters that maximize Ir utilization. At this scale, however, changing cluster size simultaneously alters structure, oxygen coverage, and the distribution of nonequivalent active sites, making it difficult to identify the physical origin of size-dependent activity. Here, we combine systematic structure sampling, density functional theory, implicit solvation, and site-resolved microkinetics to resolve OER thermodynamics across Ir_nO_x (n = 1–30) clusters supported on anatase TiO_2(101). The resulting activity landscape is strongly heterogeneous and nonmonotonic with cluster size, with substantial activity variations even among sites on the same cluster. This heterogeneity can be traced to the local oxygen coordination of the active Ir–*O motif. Increasing oxo-proximal coordination, quantified by $CN_{O,co}(Ir^*)$, raises the *OH-to-*O free-energy cost while lowering that of *O-to-*OOH formation, thereby controlling the energetic balance between the two steps surrounding *O. Electronic-structure analysis associates this coordination dependence with charge redistribution at the binding Ir center, changes in Ir–O bonding, and reorganization of the Ir 5d–O 2p electronic structure. Steady-state microkinetics preserves the thermodynamic activity hierarchy, with ensemble rates governed primarily by the most active sites at each cluster size. Single-Ir dissolution thermodynamics further shows that higher local oxygen coordination is associated with greater resistance to Ir loss, extending the significance of local coordination from OER energetics to dissolution stability. These results establish local-environment populations as the structural origin of size-dependent catalytic behavior in subnanometer IrO_x clusters, revealing local coordination rather than cluster size alone as the key structural variable underlying activity and stability.

Keywords: acidic oxygen evolution reaction; iridium oxide; subnanometer clusters; local coordination; microkinetic modeling

1 Introduction

Proton-exchange-membrane (PEM) water electrolysis is an important technology for hydrogen production, but its broader deployment is constrained by the reliance of the acidic oxygen evolution reaction (OER) on scarce iridium. [1] The challenge is therefore to reduce Ir loading while maintaining high catalytic activity and stability under strongly acidic operating conditions. [2–4] Dispersing Ir into supported single atoms or subnanometer oxide clusters provides a direct strategy to increase Ir utilization by increasing the fraction of catalytically accessible Ir atoms.[5, 6] Recent experiments on TiO_2-supported sub-nm IrO_x clusters further demonstrate the promise of supported IrO_x nanostructures for acidic OER, while spatially correlated Ir atoms incorporated into TiO_2 directly show that flexible Ir–O–Ir connectivity can tune intermediate adsorption. [7, 8] However, as Ir is confined to the subnanometer regime, changing cluster size simultaneously alters the atomic structures and local environments of the exposed catalytic sites. How these size-dependent structural changes translate into OER activity remains poorly understood.

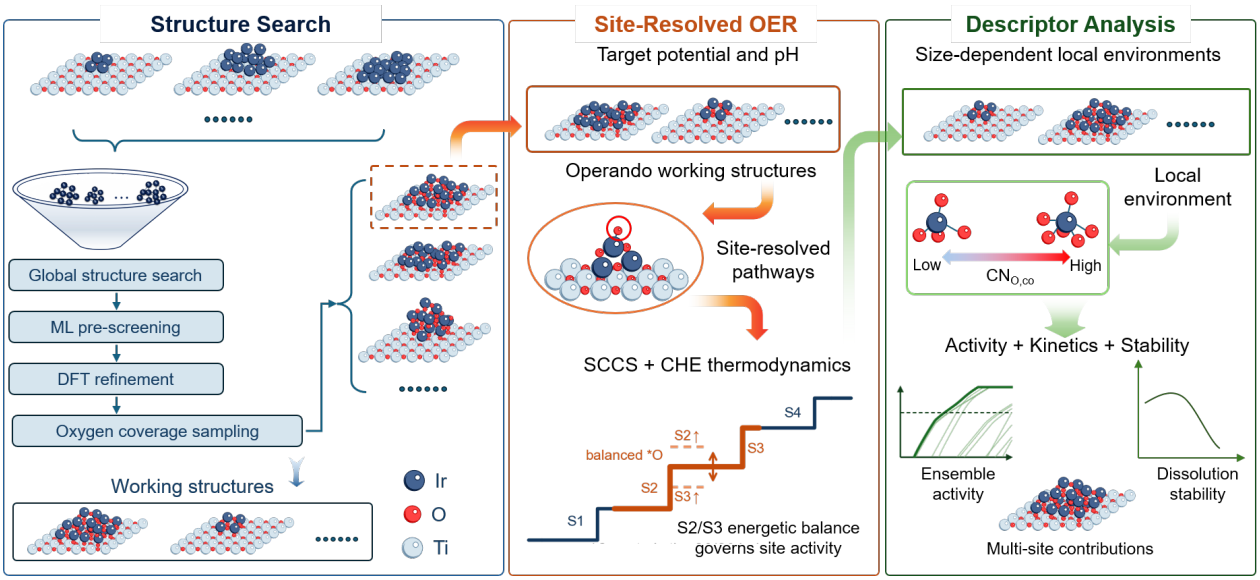


Figure 1. Computational framework for resolving structure-dependent OER activity and stability in supported Ir_nO_x clusters. Left: supported Ir clusters were generated by global structure search, refined by DFT, and sampled over oxygen coverage to determine structures under the target electrochemical conditions. Center: site-resolved OER thermodynamics were evaluated using SCCS and the computational hydrogen electrode. Right: local coordination environments were analyzed in relation to OER activity, multi-site kinetics, and single-Ir dissolution stability.

A central difficulty is that cluster size is not itself a unique structural variable. Changing the number of metal atoms simultaneously alters cluster geometry, oxygen content and oxidation state, metal–support bonding, and the distribution of nonequivalent catalytic sites.[9] Subnanometer clusters therefore need not exhibit a monotonic relationship between atom number and reactivity. [10–12] This complexity is particularly relevant to acidic OER, where local coordination can redistribute the free-energy costs among correlated oxygenated intermediates. [13–17] Operando studies further show that active structures and oxidation states can evolve under bias. [18–20] Thus, apparent size effects may reflect changes in the populations of local catalytic environments rather than a direct dependence of reactivity on atom count. Establishing this picture requires a chemically interpretable local descriptor that remains predictive across cluster sizes and connects structure to reaction energetics. Local coordination descriptors provide a natural means of resolving site heterogeneity, [21–24] but their predictive value must be distinguished from correlations inherited from cluster size or global stoichiometry. Recent approaches combining realistic structural sampling with site-resolved kinetics and interpretable local descriptors provide a basis for such analysis.[25, 26] Multiple nonequivalent sites may also contribute simultaneously to the catalytic response, [27] so a thermodynamic optimum at a single site need not determine the kinetically relevant cluster size. The remaining questions are how local environments govern site-specific OER energetics, how their populations evolve with cluster size, and whether the resulting activity trends persist under multi-site kinetics and constraints from Ir dissolution. [28]

Here, we investigate how cluster size gives rise to site-resolved OER activity and dissolution stability in Ir_nO_x (n = 1–30) clusters supported on anatase TiO_2(101). Systematic structure searches are combined with site-resolved acidic OER thermodynamics, electronic-structure analysis, and steady-state microkinetics to resolve the nonequivalent catalytic environments accessible across this size range. The resulting activity landscape is strongly heterogeneous and nonmonotonic, showing that cluster size alone does not specify the thermochemistry of an individual active site. Instead, we identify the local oxygen coordination around the active Ir–*O motif, quantified by $CN_{O,co}(Ir^*)$, as the dominant structural variable associated with this heterogeneity. This coordination modulates the energetic balance between *OH oxidation to *O and subsequent *OOH formation, while its population varies systematically across cluster sizes. Electronic-structure analysis further reveals the accompanying changes in local charge distribution and Ir–O bonding, providing a microscopic basis for the coordination dependence of the reaction energetics. Steady-state microkinetics confirms that these site-level thermodynamic trends persist when contributions from multiple sites are considered. We further examine single-Ir dissolution thermodynamics and find that local oxygen coordination is also associated with

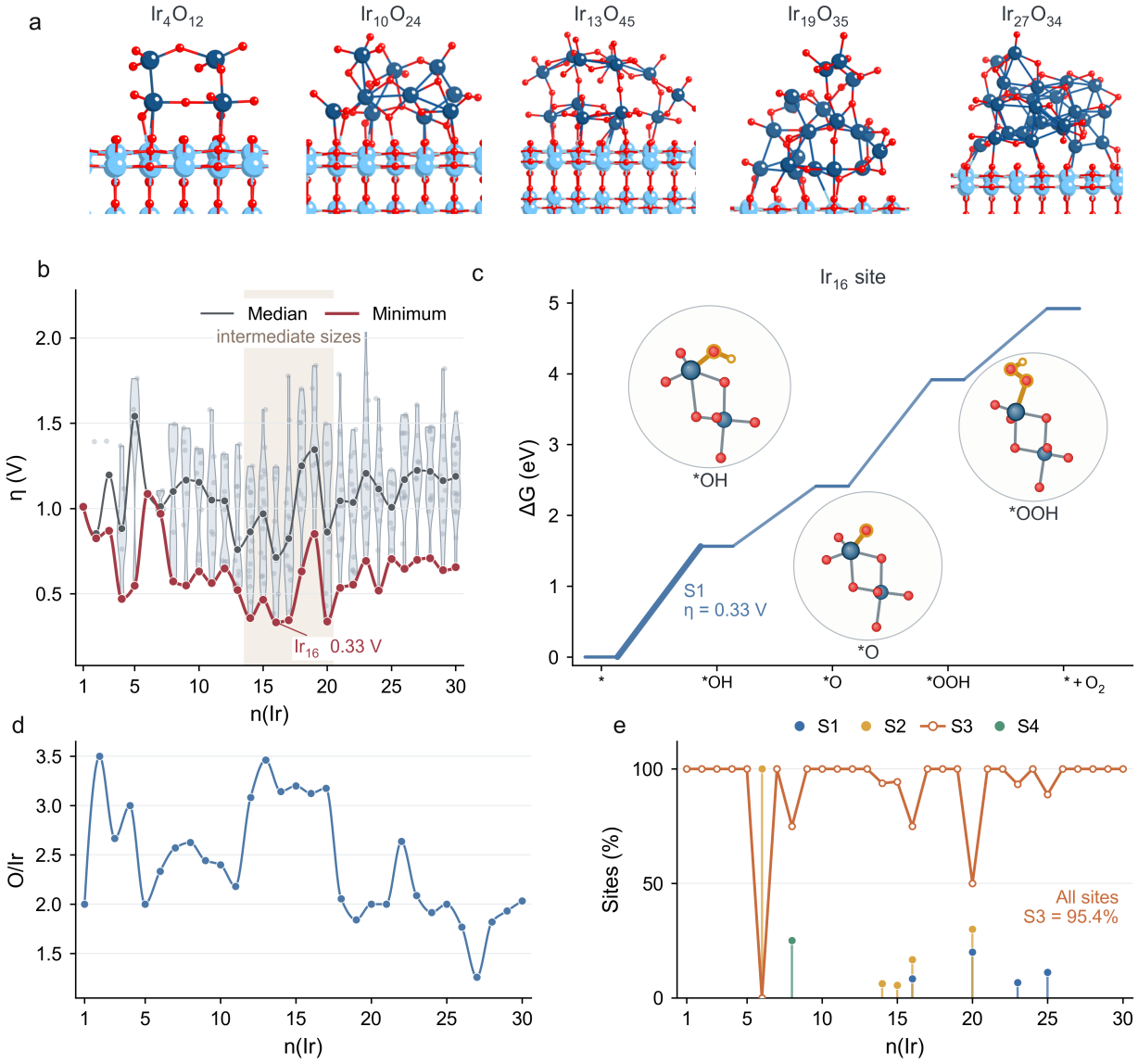


Figure 2. Working structures and site-resolved OER thermodynamics. (a) Representative DFT-selected $Ir_nO_x/TiO_2(101)$ working structures across the investigated cluster-size range. (b) Thermodynamic OER overpotentials for the 303 complete SCCS pathways. Points represent individual sites; violin distributions are shown for sizes with at least four complete pathways, and lines indicate the median and minimum overpotential at each size. (c) Cumulative free-energy profile of the lowest-overpotential Ir_{16} site at $U = 0$ V versus RHE and pH = 0, together with the relaxed local structures of the *OH, *O, and *OOH intermediates. (d) O/Ir ratio of the selected working structure as a function of cluster size. (e) Distribution of the potential-determining step (PDS) across cluster sizes. The line shows the fraction of S3-limited pathways at each size, while markers indicate nonzero fractions of S1-, S2-, and S4-limited pathways. S1–S4 denote the four AEM proton–electron transfer steps defined in the text; S3 is potential determining for 95.4% of all pathways.

resistance to Ir loss. Together, these results establish local-environment populations as the structural basis of size-dependent behavior in supported subnanometer IrO_x. Cluster size determines the distribution of local coordination environments that govern OER energetics and are also associated with resistance to Ir dissolution.

2 Results and Discussion

Figure 1 summarizes the computational framework used to resolve the size-dependent catalytic behavior of supported Ir_nO_x clusters. The analysis begins with systematic sampling of the supported Ir framework and oxygen configurations to determine the structures under the chosen electrochemical conditions. OER thermodynamics were then evaluated for each accessible active site. Structural and electronic analyses examine the local environments underlying the resulting activity variations, while steady-state microkinetics evaluates their consequences for the catalytic response of each cluster. Finally, single-Ir dissolution thermodynamics is used to assess the stability of the sampled local environments against Ir loss.

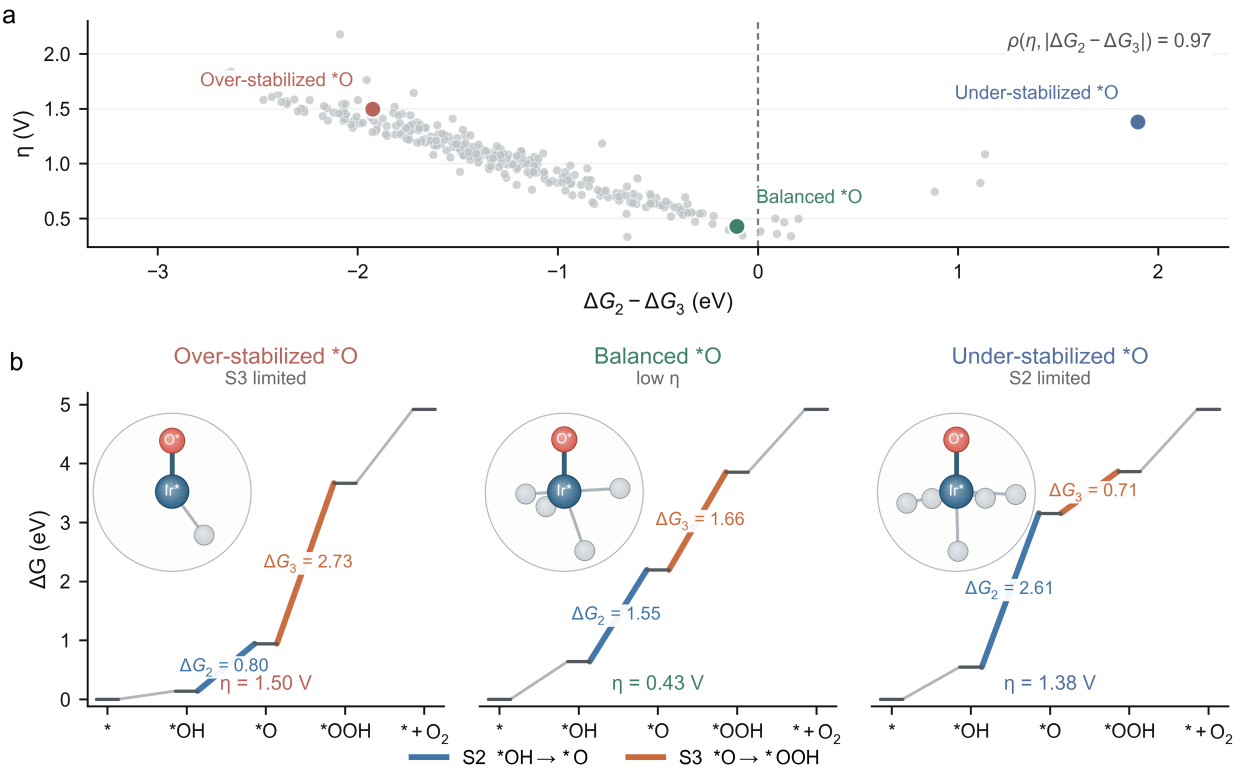


Figure 3. Thermodynamic balance of the *O intermediate. (a) Thermodynamic overpotential of all 303 complete SCCS pathways as a function of the signed balance coordinate $D_O = \Delta G_2 - \Delta G_3$. Negative and positive values correspond to relative over- and under-stabilization of *O, respectively, and the dashed line marks $\Delta G_2 = \Delta G_3$. Colored points identify the three Ir_{20} sites shown in panel b. (b) Cumulative free-energy profiles at $U = 0$ V versus RHE and pH = 0 for three sites on the same Ir_{20} working structure, representing over-stabilized, balanced, and under-stabilized *O energetics. S2 (*OH → *O) and S3 (*O → *OOH) are highlighted in blue and orange, respectively, with S1 and S4 shown in gray. Insets show the corresponding local *O environments.

2.1 Size-dependent OER activity of supported Ir_nO_x clusters

For subnanometer IrO_x clusters, a given Ir atom count can accommodate multiple metal–support geometries and oxygen stoichiometries. We therefore used a two-stage configurational search, first identifying low-energy supported Ir skeletons and then sampling oxygen coverage and O-atom configurations on the selected motifs (Figure 1). Related approaches have been used to account for coverage- and stoichiometry-dependent structures under reactive conditions. [29–31]

Supported Ir_1–Ir_{30} candidates on anatase $TiO_2(101)$ were generated using AGOX with the UMA machine-learning potential. [32–35] This range spans isolated Ir species to clusters comparable to atomistic models of experimentally characterized sub-nm IrO_x/TiO_2 catalysts. [7] Structurally distinct low-energy candidates were then optimized and reranked at the PBE-D3 level[36]. Among 55 metallic motifs, reranking changed the lowest-energy structure for seven cluster sizes, showing that UMA does not always preserve the energetic ordering of competing motifs. The lowest-energy PBE-D3 motif for each size was retained for the oxygen-coverage search.

Sampling of oxygen stoichiometry and O-atom configurations yielded 277 oxygen-covered structures for PBE-D3 evaluation. Their grand potentials were evaluated at U = 1.5 V, pH = 0, and 298 K, and the lowest-grand-potential structure was selected for each Ir_n. The selected $Ir_nO_x/TiO_2(101)$ structures exhibit substantial size-dependent variation in both cluster geometry and oxygen stoichiometry (Figure 2a).

For each exposed Ir center, we evaluated the four proton–electron transfers along the adsorbate evolution mechanism (AEM), a widely used mechanistic pathway for acidic OER on Ir-based catalysts[7, 37]: S1, * → *OH; S2, *OH → *O; S3, *O → *OOH; and S4, *OOH → * + O_2. The potential-determining step (PDS) was identified as the step with the largest free-energy change at U = 0 V, from which the thermodynamic overpotential was obtained.

Figure 2b reveals a strongly heterogeneous and nonmonotonic activity landscape across Ir_nO_x clusters. The 303 complete pathways span overpotentials of 0.33–2.18 V, with the lowest values found for Ir_{16} (0.33 V), Ir_{20} (0.34 V), Ir_{17} (0.35 V), and Ir_{14} (0.36 V). Neither the minimum nor median overpotential varies monotonically with Ir atom count, and substantial activity variations also occur among sites of the same size. Cluster size alone therefore does not determine site activity, motivating examination of the local environments underlying this heterogeneity. The lowest-overpotential pathway on Ir_{16} is shown in Figure 2c, with η = 0.33 V and S1 as the PDS.

Overall oxygen loading does not account for the activity trend. The O/Ir ratio also varies nonmonotonically with cluster size but does not track the site-resolved overpotentials (Figure 2d). Thus, neither Ir atom count nor overall stoichiometry captures the observed thermodynamic heterogeneity. Additional analysis of the selected oxygen-covered structures is provided in Supporting Information, Section S1 and Figure S1.

Despite the broad activity distribution, S3 ($^*O \rightarrow {}^*OOH$) is potential determining for 95.4% of the 303 pathways (Figure 2e). This assignment is consistent with the experimentally supported AEM for TiO_2-supported sub-nm IrO_x clusters and the corresponding identification of $^*O$-to-$^*OOH$ formation as the rate-determining step. [7] The activity variation therefore occurs predominantly within the same S3-limited thermodynamic regime rather than through changes in the limiting step. Implicit solvation preserves the dominant PDS and site ranking (Supporting Information, Section S4).

2.2 Local energetic origin of site-dependent OER activity

Because $^*O$ is shared by the S2 and S3 steps, its relative stability directly redistributes the free-energy cost between $^*OH$ oxidation and $^*OOH$ formation. We quantify this balance using the signed coordinate $D_O = \Delta G_2 - \Delta G_3$ (Figure 3a). Negative D_O corresponds to relative over-stabilization of $^*O$, for which S2 is favorable but S3 is penalized, whereas positive D_O reflects under-stabilization of $^*O$ and the opposite energetic imbalance. Across all 303 complete pathways, η increases strongly with $|D_O|$ (Spearman ρ = 0.967), and the lowest-overpotential sites are concentrated near $\Delta G_2 \approx \Delta G_3$. Low OER overpotential is therefore favored by a balance between the free-energy costs of forming $^*O$ (S2) and converting it to $^*OOH$ (S3), rather than by preferential stabilization of $^*O$ itself.

Figure 3b illustrates the same energetic trade-off among three sites on the Ir_{20} supported cluster as an example. At the over-stabilized-$^*O$ site, $(\Delta G_2, \Delta G_3) = (0.80, 2.73)$ eV, resulting in a large S3 penalty and η = 1.50 V. At the balanced site, the two steps become comparable, with $(\Delta G_2, \Delta G_3) = (1.55, 1.66)$ eV and η = 0.43 V. Further destabilization of $^*O$ reverses the imbalance: $(\Delta G_2, \Delta G_3) = (2.61, 0.71)$ eV, making S2 potential determining and increasing η to 1.38 V. This picture is consistent with the experimentally motivated view that weakening $^*O$ binding promotes OER on supported IrO_x[7].

The energetic balance identified in Figure 3 raises the question of which local structural features regulate the relative stability of $^*O$ across nonequivalent sites. We therefore examined a set of structural quantities

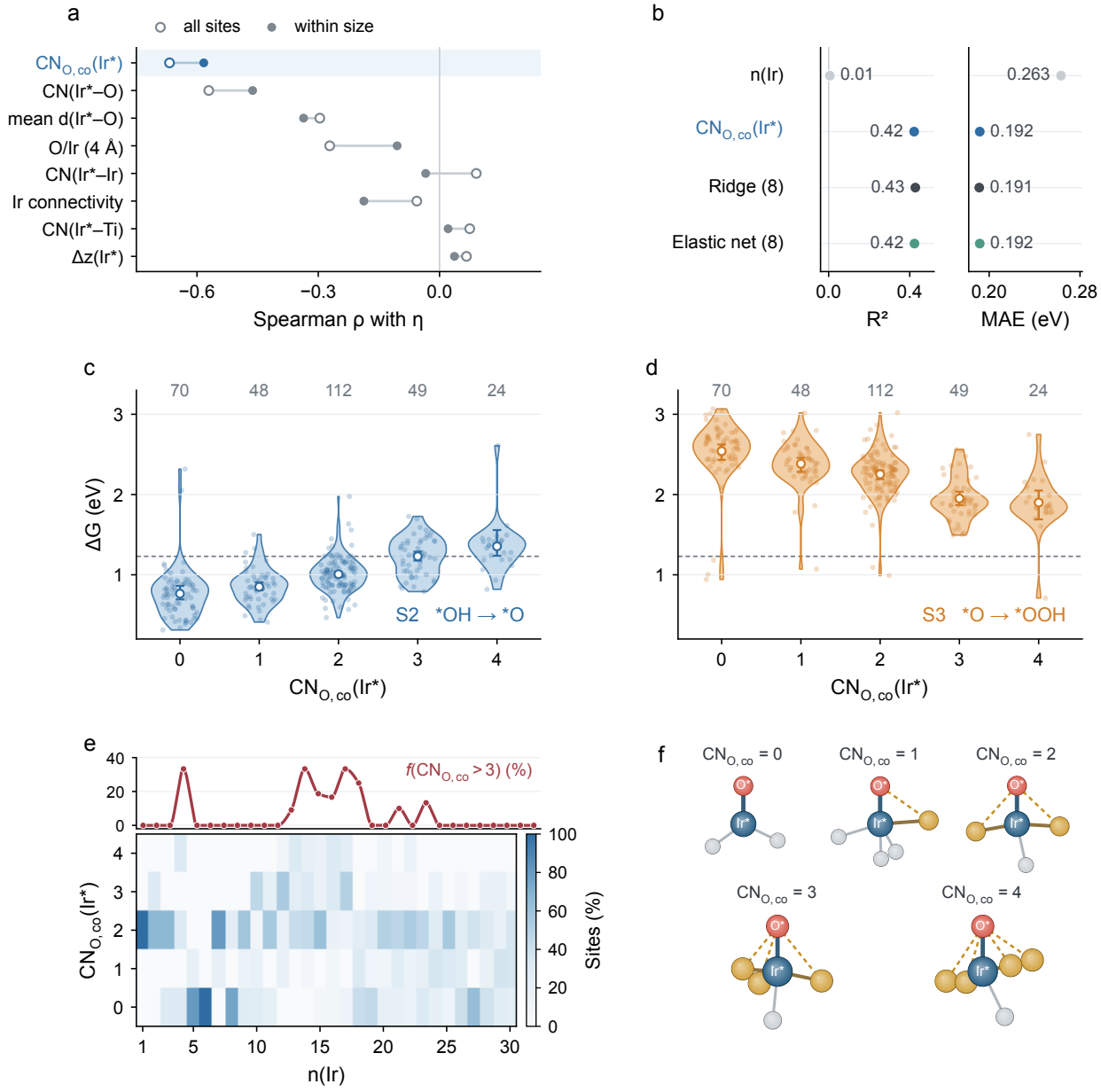


Figure 4. Oxo-proximal coordination and OER thermodynamics. (a) Overall and size-centered Spearman correlations between η and eight structural quantities. (b) Leave-one-size-out prediction of η using Ir atom count, $CN_{O,co}(Ir^*)$, and models incorporating all eight structural quantities, evaluated by R^2 and MAE. (c,d) Distributions of the S2 and S3 free-energy changes as a function of $CN_{O,co}(Ir^*)$. Points represent individual sites, violins show the distributions, and open circles and error bars indicate the means and 95% size-resampled confidence intervals. The dashed line marks 1.23 eV, and numbers indicate the number of sites in each group. (e) Size-dependent distribution of $CN_{O,co}(Ir^*)$ across the sampled active sites; the upper trace shows the fraction of sites with $CN_{O,co}(Ir^*) > 3$. (f) Representative relaxed local environments with $CN_{O,co}(Ir^*) = 0$–4. Ir* denotes the Ir center bound to O*. Counted O co-ligands satisfy both Ir*–O < 2.5 Å and O*–O < 2.9 Å, with O* itself excluded.

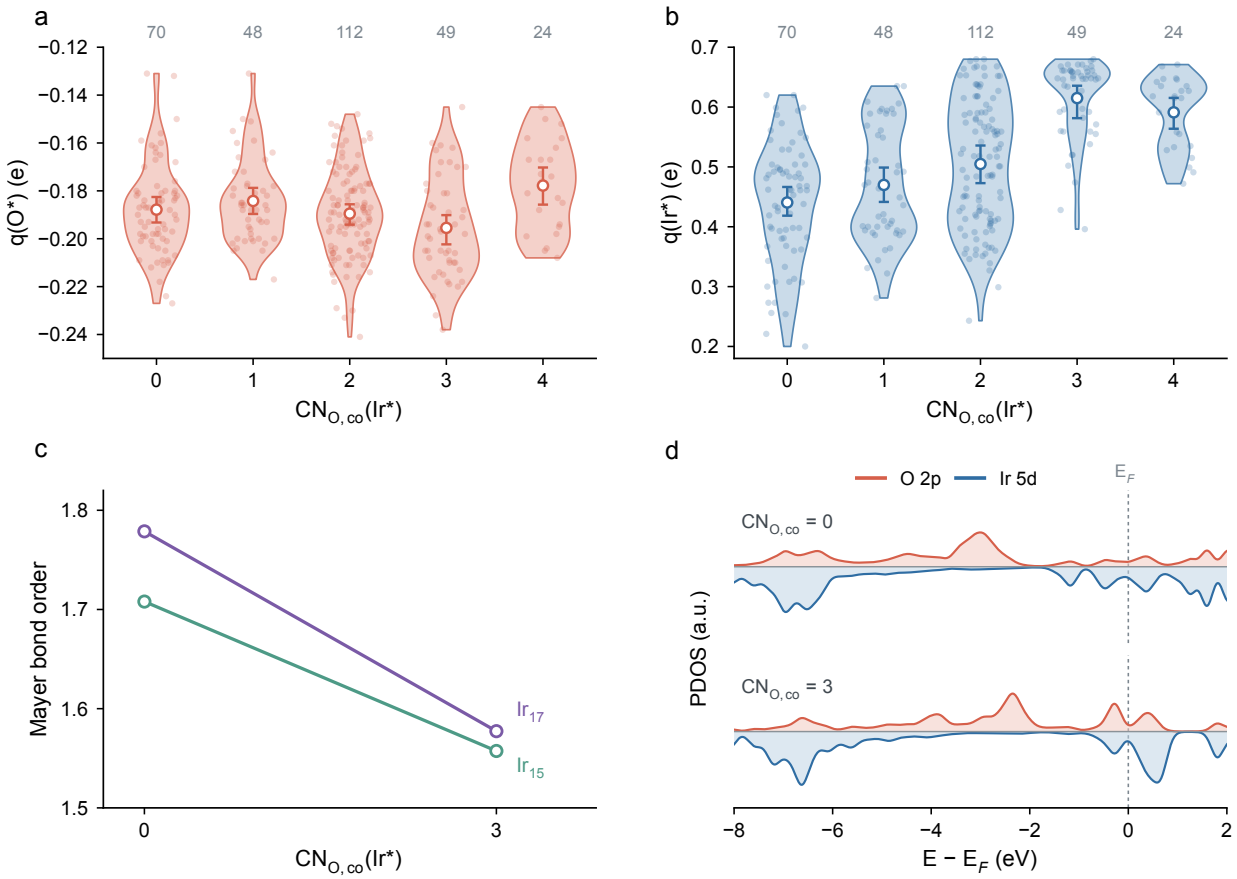


Figure 5. Electronic response of the active Ir–O unit. (a,b) Hirshfeld charge distributions of the active O* and binding Ir*, respectively, for all 303 sites grouped by $CN_{O,co}(Ir^*)$. Points represent individual sites, violins show the distributions, and open circles and error bars indicate the means and 95% size-resampled confidence intervals. Numbers indicate the number of sites in each group. The overall and size-centered Spearman correlations are −0.032 and −0.166 for O*, and 0.516 and 0.217 for Ir*, respectively. (c) Mayer bond orders of the active Ir*–O* pairs for representative $CN_{O,co}(Ir^*) = 0$ and 3 sites on Ir_{15} and Ir_{17}; lines connect sites within the same working structure. (d) Active-O 2p and binding-Ir 5d projected densities of states (PDOS) for $CN_{O,co}(Ir^*) = 0$ and 3 sites on the same Ir_{17} working structure. Energies are referenced to the Fermi level, and the Ir 5d PDOS is mirrored for clarity. All spectra are shown on the same scale using a Gaussian broadening of 0.12 eV.

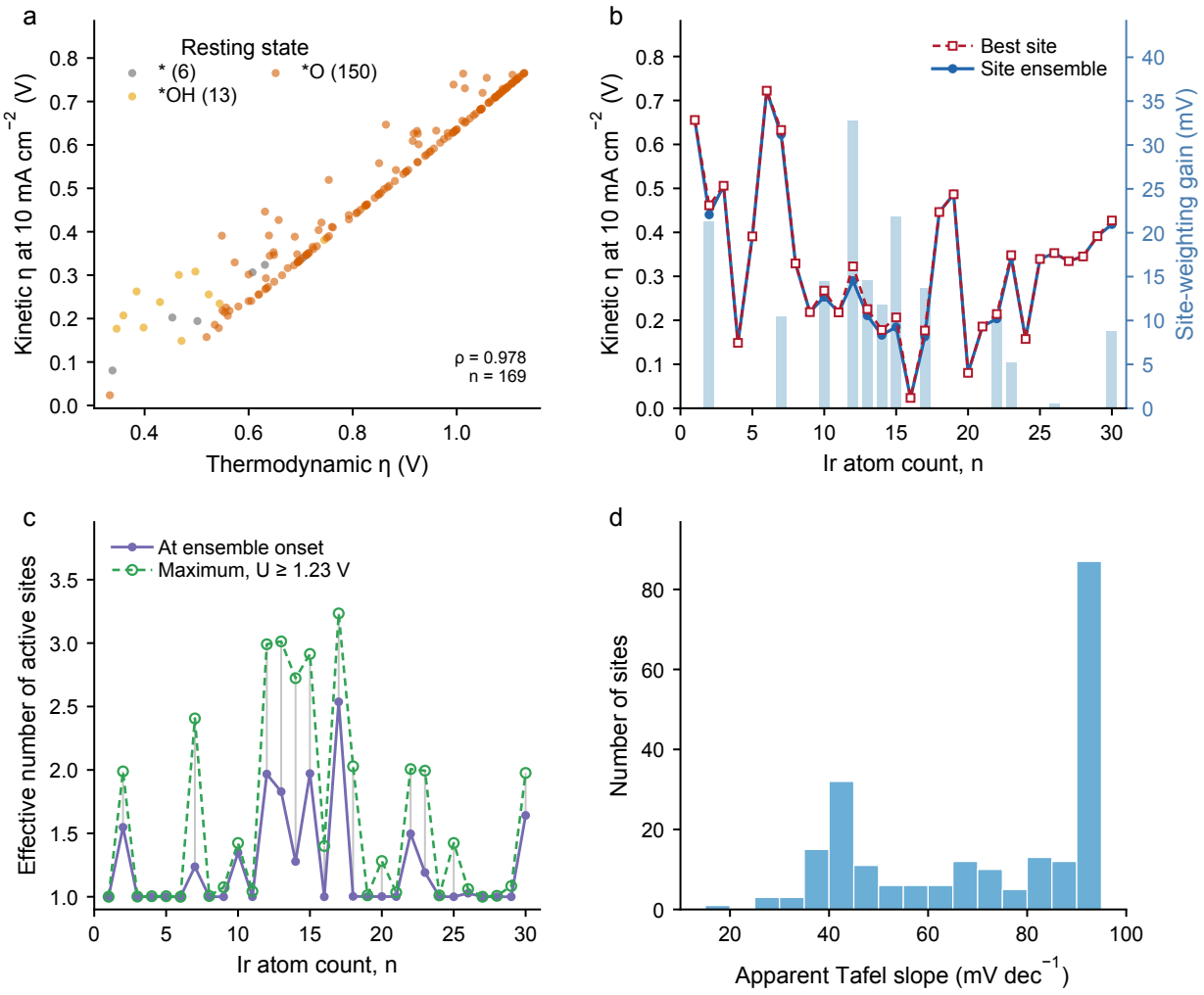


Figure 6. Site-resolved steady-state microkinetics. (a) Thermodynamic versus kinetic overpotential for sites reaching 10 mA cm^{-2} within the sampled potential range, colored by the dominant resting state at the corresponding potential. (b) Best-site and site-ensemble kinetic overpotentials for each cluster size (left axis); bars show the reduction in onset overpotential upon summing the contributions from all sampled sites (right axis). (c) Effective number of rate-contributing sites at the ensemble onset and the maximum value reached above 1.23 V. (d) Distribution of apparent Tafel slopes fitted over 0.1–10 mA cm^{-2}.

(Supporting Information, Section 5) describing the active Ir–*O environment, Ir connectivity, and interaction with the support. Figure 4a compares their associations with η across all sites and after removal of the corresponding size-dependent mean. Among the nonconstant quantities, $CN_{O,co}(Ir^*)$ shows the strongest monotonic relationship with activity, which quantifies the local oxygen coordination surrounding the active Ir–O* unit by counting O co-ligands that are coordinated to Ir* and located in proximity to O*. Its Spearman correlation with η is −0.668 across all 303 sites, showing that $CN_{O,co}(Ir^*)$ captures site-to-site activity variation beyond the differences between cluster sizes.

Leave-one-size-out cross-validation further supports the transferability of $CN_{O,co}(Ir^*)$ across cluster sizes (Figure 4b). Ir atom count alone has essentially no predictive power ($R^2 = 0.01$), whereas $CN_{O,co}(Ir^*)$ reaches $R^2 = 0.42$. Including all structural descriptors yields only a marginal improvement ($R^2 = 0.43$), indicating that $CN_{O,co}(Ir^*)$ captures most of the transferable structural information relevant to OER activity. Additional validation results are provided in Supporting Information, Section S5. More importantly, $CN_{O,co}(Ir^*)$ directly relates local structure to the S2/S3 energetic balance. Higher $CN_{O,co}(Ir^*)$ increases the *OH-to-*O free-energy change (S2, $\rho = +0.641$) while decreasing the *O-to-*OOH free-energy change (S3, $\rho = -0.655$) (Figure 4c-d). Increasing oxo-proximal coordination therefore shifts the energetic penalty from S3 toward S2, providing a structural basis for the balance identified in Figure 3.

The population analysis in Figure 4e connects the site-level relationship to cluster size. The distribution of $CN_{O,co}(Ir^*)$ varies strongly and nonmonotonically across Ir_nO_x, with highly coordinated sites ($CN_{O,co}(Ir^*) > 3$) enriched in the intermediate-size regime. Their fraction averages 14.8% for Ir_{14}–Ir_{20}, compared with 3.9% outside this range. Cluster size therefore modulates the population of local Ir–oxo environments rather than prescribing a unique active-site structure, providing a population-level explanation for the variation in OER activity across cluster sizes.

2.3 Electronic response of the active Ir–O Unit

The coordination dependence of the S2 and S3 energetics motivates an examination of how the local oxygen environment modifies the electronic structure of the active Ir–O unit. We therefore analyzed charge redistribution, Ir–O bond order, and the local Ir 5d–O 2p electronic states. The Hirshfeld charge of O* shows little systematic dependence on $CN_{O,co}(Ir^*)$ over the populated range of 0–4 (Figure 5a). In contrast, the binding Ir* becomes more positive with increasing local oxygen coordination (Figure 5b). Its mean Hirshfeld charge increases from +0.441 e at $CN_{O,co}(Ir^*) = 0$ to +0.615 e and +0.591 e at $CN_{O,co}(Ir^*) = 3$ and 4, respectively, with an overall correlation of $\rho = 0.516$. These results indicate that the coordination dependence cannot be described by changes in the O* charge alone, but involves polarization of the binding Ir center. This emphasis on an Ir-centered response is consistent with experimental reports that flexible Ir–O–Ir coordination and bias-driven local charge accumulation can tune the oxidation state and intermediate adsorption of oxide-supported Ir sites.[8, 20]

A more direct site-level comparison is provided by the Ir*–O* bond order. As shown in Figure 5c, the Mayer bond order decreases from 1.708 to 1.557 for Ir_{15} and from 1.779 to 1.577 for Ir_{17} as $CN_{O,co}(Ir^*)$ increases from 0 to 3. The consistent decrease within each structure indicates that higher local oxygen coordination is associated with a lower Ir*–O* bond order. The Ir_{17} PDOS provides an orbital-resolved comparison of the same local environments (Figure 5d). From $CN_{O,co}(Ir^*) = 0$ to $CN_{O,co}(Ir^*) = 3$, the integrated O 2p weight within 1 eV of the Fermi level increases from 0.292 to 0.600, while the corresponding Ir 5d weight increases from 0.790 to 1.190. Their normalized PDOS overlap over −6 to 1 eV also increases modestly, from 0.476 to 0.509.

Taken together, these analyses show that the structural dependence identified by $CN_{O,co}(Ir^*)$ cannot be reduced to a single charge or bonding metric. Instead, changes in the local oxygen environment are accompanied by coupled electronic responses involving polarization of the binding Ir center, changes in Ir–O bond order, and redistribution of Ir 5d–O 2p states. We therefore interpret $CN_{O,co}(Ir^*)$ as a geometric descriptor of a broader electronic reorganization of the active Ir–O unit.

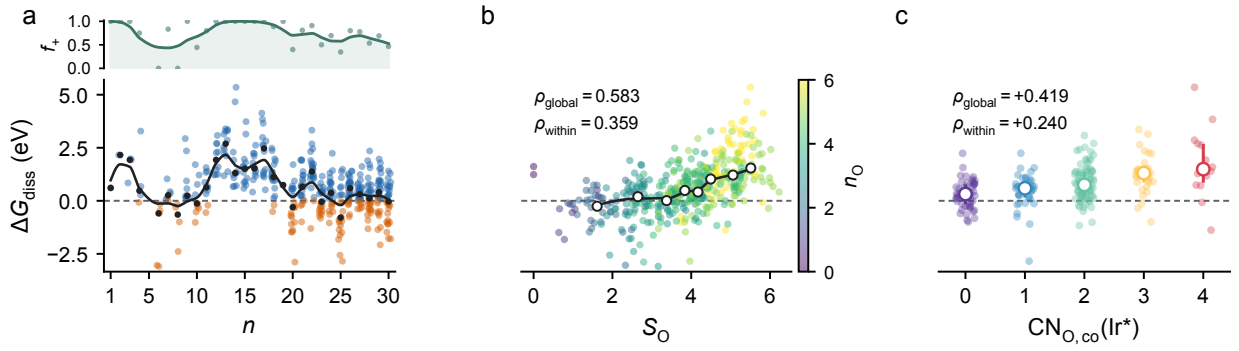


Figure 7. Local coordination and Ir dissolution stability at pH = 0 and 1.23 V versus RHE. (a) Site-resolved dissolution free energies by cluster size. Blue and orange denote positive and negative values, respectively; black markers show the size medians, and the upper trace gives the fraction f_+ with $\Delta G_{diss} > 0$. (b) ΔG_{diss} as a function of the distance-weighted O-coordination measure, colored by the O coordination number of the removed Ir. Open markers show equal-population bin medians. (c) ΔG_{diss} as a function of $CN_{O,co}(Ir^*)$ for 215 activity–stability matched sites. Open markers and bars summarize the median and interquartile range. The reported Spearman coefficients distinguish global from within-parent associations.

2.4 Steady-state kinetics and multi-site contributions

To assess whether the thermodynamic activity trends persist under steady-state conditions, we evaluated all complete pathways using a four-state reversible microkinetic model with potential-dependent activation barriers following Mao *et al.*.[38] Of the 303 pathways, 169 reach the reference current density within the sampled potential range. Their thermodynamic and kinetic overpotentials are strongly correlated (ρ = 0.978; Figure 6a), showing that the site-activity ranking is largely preserved under steady-state kinetics. *O is the dominant resting state for 150 of these sites, consistent with the importance of the S2/S3 energetics identified above.

Combining contributions from multiple sites produces only a modest change in the kinetic response (Figure 6b,c). Relative to the best-site response, the ensemble onset decreases by 5.5 mV on average and by at most 32.7 mV for Ir_{12}. The effective number of contributing sites remains close to unity near onset for most cluster sizes, indicating that the kinetic response is dominated by the most active local environments. At higher potentials, contributions from multiple sites become more appreciable. The apparent Tafel slopes are broadly distributed, with a median of 80.3 mV dec^{-1} (Figure 6d), reflecting site-specific differences within the common AEM framework.

Furthermore, implicit solvation modifies the quantitative kinetic onsets but does not alter the resulting activity hierarchy. Repeating the microkinetic analysis using the corresponding vacuum free-energy profiles gives nearly identical site rankings to the SCCS results (ρ = 0.979), while Ir_{16} remains the most active cluster size. The median change in kinetic overpotential upon inclusion of SCCS is only −7.3 mV, and the Tafel-slope distribution is essentially unchanged. These results are consistent with the substantial cancellation of solvation contributions between adjacent intermediates and show that implicit solvation refines the quantitative energetics without changing the principal site- and size-dependent kinetic trends. Additional solvation comparisons are provided in Supporting Information, Section S4.

The persistence of the activity landscape was further examined with respect to the assumed kinetic parameters. Across the lower intrinsic-barrier range, the kinetic rankings remain strongly correlated with the baseline model, while the most active cluster can switch between Ir_{16} and Ir_{20}. At larger barriers, progressively fewer sites reach the reference current within the sampled potential range. The microkinetic analysis therefore supports a robust intermediate-size activity regime rather than a unique, barrier-independent optimum. The full barrier and transfer-coefficient sensitivity analysis is reported in Supporting Information, Section S7, Figure S6, and Table S10.

2.5 Local coordination and dissolution stability

An important question is whether the local environments associated with favorable OER thermodynamics also resist Ir loss under acidic operating conditions.[39, 40] We therefore examined whether the coordination trends identified above extend to dissolution stability by comparing each accessible Ir center with its relaxed single-Ir vacancy, taking aqueous Ir^{3+} as the dissolved reference species. This choice is consistent with the Ir^{3+} stability domain predicted for strongly acidic solution in the Ir Pourbaix diagram and with recent measurements of the pH dependence of Ir dissolution. [41] Positive ΔG_{diss} indicates that the intact structure is thermodynamically favored against this defined dissolution channel. At pH 0 and 1.23 V versus RHE, 293 of the 413 structurally audited sites exhibit positive ΔG_{diss}. The broad distributions within individual cluster sizes and the nonmonotonic variation of their medians also show that dissolution resistance is governed primarily by the local environment of the Ir center rather than by cluster size alone.

The site-resolved distributions are summarized in Figure 7a. Local oxygen coordination provides an initial structural trend in the dissolution thermodynamics. A distance-weighted O-coordination measure is positively correlated with ΔG_{diss} both globally ($\rho = +0.583$) and within parent structures ($\rho = +0.359$; Figure 7b), indicating greater resistance to Ir loss in more highly coordinated environments. More importantly, the oxo-proximal coordination descriptor $\mathrm{CN_{O,co}}(\mathrm{Ir^*})$ identified from the OER thermodynamics exhibits the same favorable tendency. Among the 215 sites shared by the activity and stability datasets, increasing $\mathrm{CN_{O,co}}(\mathrm{Ir^*})$ is associated with higher ΔG_{diss} ($\rho = +0.419$ globally and $+0.240$ within parent structures) and lower OER overpotential ($\rho = -0.710$ and -0.611, respectively; Figure 7c). Thus, the local coordination environment associated with favorable OER energetics is also associated with greater thermodynamic resistance to Ir dissolution.

This common structural dependence does not imply a direct relationship between activity and stability: the within-parent correlation between η and ΔG_{diss} is essentially zero ($\rho = -0.012$). Rather, $\mathrm{CN_{O,co}}(\mathrm{Ir^*})$ provides a structural connection between the two properties, with higher oxo-proximal coordination associated with both lower overpotential and greater dissolution resistance.

It should be mentioned that the calculated ΔG_{diss} describes the thermodynamics of the defined single-Ir dissolution pathway and does not represent a kinetic barrier or catalyst lifetime. A joint activity–stability analysis identifies five Pareto-optimal sites within the sampled population, demonstrating that favorable combinations of OER activity and dissolution stability are accessible (Figure S8). Additional methodological details and stability analyses are provided in Supporting Information, Section S8.

3 Conclusion

In summary, our analysis of $\mathrm{Ir}_n\mathrm{O}_x$ (n = 1–30) on anatase $\mathrm{TiO_2}(101)$ establishes that cluster size influences OER activity by modulating the populations of distinct local catalytic environments. Site-resolved energetics vary substantially within and across cluster sizes, with the oxo-proximal coordination $\mathrm{CN_{O,co}}(\mathrm{Ir^*})$ emerging as the dominant structural variable underlying this heterogeneity. Increasing $\mathrm{CN_{O,co}}(\mathrm{Ir^*})$ systematically shifts the free-energy penalty from *O oxidation to *O formation, such that the most favorable sites approach an energetic balance between S2 and S3. Electronic-structure analysis further associates this behavior with changes in the charge, bonding, and Ir 5d–O 2p electronic structure of the active Ir–O unit. Steady-state microkinetics preserves the thermodynamic site-activity hierarchy after accounting for intermediate populations, reaction reversibility, and simultaneous turnover at multiple sites.

The same structural analysis also provides insight into dissolution stability. More highly oxygen-coordinated Ir centers are generally more resistant to Ir loss, and $\mathrm{CN_{O,co}}(\mathrm{Ir^*})$ is favorably associated with both low OER overpotential and high dissolution free energy. At the same time, the absence of a direct correlation between activity and stability shows that one is not simply a consequence of the other. Local coordination instead emerges as a shared structural dimension along which favorable catalytic energetics and dissolution resistance can coexist. These findings suggest that the performance of subnanometer IrO_x catalysts is better understood in terms of the local environments realized within a cluster than by its size alone, and motivate the use of local coordination as a design criterion for balancing Ir utilization, OER activity, and stability.

4 Computational Methods

4.1 Model system and structural sampling

Ir_n clusters with n = 1–30 were supported on a 2 × 2 anatase TiO_2(101) slab containing three TiO_2 trilayers ($O_{384}Ti_{192}$). The periodic cell dimensions were 30.26 × 20.66 × 28.48 Å. Structural sampling was performed in two stages to separately explore the supported Ir skeleton and its oxygen coverage. Supported Ir_n/TiO_2 structures were first generated using AGOX with random and rattle structure generators within an 8 Å confinement region above the anatase surface. [32–34] UMA machine learning potential was used for candidate evaluation and local pre-relaxation.[35]

Structurally similar metallic candidates were removed using a permutation-aware Ir-centered fingerprint constructed from Ir–Ir distances, nearest Ir–O and Ir–Ti distances, and Ir heights above the support. Candidates within 0.5 eV of the lowest UMA energy for each size were retained for DFT reranking, and the lowest-energy DFT structure obtained for each cluster size was used as the metallic reference for subsequent oxygen sampling. UMA was used only for structure generation and prescreening; all reported structural selections and reaction energetics were based on DFT calculations.

Oxygen coverage and configuration were subsequently sampled on each selected supported Ir skeleton using chemistry-guided O placement, addition, removal, and structural perturbation followed by UMA relaxation. Candidate structures were required to form a connected oxide component and to avoid peroxo configurations. Structurally distinct configurations were then evaluated by DFT, yielding 269 oxygen-coverage/isomer states for working-state selection. Supporting Information, Section S1 reports the metallic reranking audit, candidate counts before and after the chemical and fingerprint gates, the targeted lower-coverage rescue, grand-potential working-state selection, and the subsequent active-site sampling inventory.

The sampled oxygen-coverage states were compared at U = 1.5 V versus RHE, pH = 0, and 298 K. For state i containing n_O non-slab oxygen atoms, the grand potential was evaluated using the DFT energy and the computational hydrogen electrode (CHE) oxygen chemical potential,

$$\Omega_i = E_i + n_O(Z_O - \mu_O). \tag{1}$$

The lowest-Ω sampled state was selected as the working structure for each cluster size.

Candidate OER centers were identified as terminal-oxo, coordinatively accessible Ir sites that satisfied the geometric requirements for monodentate *OOH formation and were separated from the cluster–support interface. Supporting Information, Section S1 and Figure S1 report the grand-potential separation from the best sampled state at another coverage, the size-resolved Boltzmann effective populations, and the thermally accessible oxygen coverages.

4.2 DFT and OER thermochemistry

All DFT calculations were performed with CP2K/Quickstep using the Gaussian and plane-wave method.[42] The PBE exchange–correlation functional,[36] GTH pseudopotentials,[43] and DZVP-MOLOPT-SR-GTH basis sets[44] were used for Ir, Ti, O, and H. The plane-wave density and relative cutoffs were 400 and 60 Ry, respectively, with Γ-point sampling. The SCF convergence threshold was 10^{-5}, and electronic occupations were treated using Fermi–Dirac smearing at 2000 K.

All clean, *OH, *O, and *OOH structures were optimized using the BFGS algorithm. The lower two TiO_2 trilayers were fixed, whereas the upper trilayer and all cluster and adsorbate atoms were relaxed. Geometry optimization was continued until the maximum and root-mean-square forces were below 4.5×10^{-4} and 3.0×10^{-4} Ha bohr^{-1}, respectively.

Solvation effects were evaluated using the self-consistent continuum solvation (SCCS) model of Andreussi *et al.* with a relative permittivity of 78.36.[45] The dielectric cavity was defined by density thresholds of $\rho_{min} = 10^{-5}$ and $\rho_{max} = 10^{-4}$ in CP2K atomic units. The solvent correction was evaluated on each vacuum-relaxed structure as

$$\Delta E_{solv} = E_{SCCS} - E_{vac}. \tag{2}$$

Supporting Information, Section S2 reports the SCCS acceptance criteria, state-level calculation inventory, complete-pathway provenance join, and the eight selected centers excluded because at least one required state was missing or failed the acceptance gate (Table S1).

The acidic AEM considered here follows the mechanism assigned from the pH-dependent experimental measurements of Chu *et al.*. [7] The four proton-coupled electron-transfer steps were represented as

$$^{*} + H_2O \rightarrow {}^{*}OH + H^{+} + e^{-}, \tag{3}$$

$$^{*}OH \rightarrow {}^{*}O + H^{+} + e^{-}, \tag{4}$$

$$^{*}O + H_2O \rightarrow {}^{*}OOH + H^{+} + e^{-}, \tag{5}$$

$$^{*}OOH \rightarrow {}^{*} + O_2 + H^{+} + e^{-}. \tag{6}$$

Free energies were evaluated within the CHE framework at pH 0 and 298.15 K.[13, 14] Zero-point-energy and entropic corrections were included in the free-energy calculations. The O_2 free energy was determined using the experimental water-formation free energy of 4.92 eV.

The thermodynamic overpotential for each complete pathway was calculated as

$$\eta = \frac{\max_i(\Delta G_i)}{e} - 1.23\ \mathrm{V}. \tag{7}$$

At an applied potential U, the free energy of each one-electron step was evaluated as

$$\Delta G_i(U) = \Delta G_i(0) - eU. \tag{8}$$

4.3 Single-Ir dissolution thermodynamics

Single-Ir dissolution was evaluated from the relaxed working $^{*}O$ structures by comparing each intact Ir site with its corresponding relaxed vacancy structure. For each target Ir atom, O atoms coordinated exclusively to that Ir were removed together with the dissolving Ir, whereas O atoms shared with neighboring Ir or bonded to the TiO_2 support were retained. Specifically, an O atom was classified as Ir-exclusive when its distance to the target Ir was below 2.5 Å and it was coordinated neither to another Ir within 2.5 Å nor to a support Ti within 2.4 Å. Depending on the local coordination environment, $m = 0$–3 Ir-exclusive O atoms accompanied removal of the target Ir. The resulting vacancy structures were relaxed with the lower two TiO_2 trilayers fixed, using the same convergence criteria as for the OER structures.

At pH 0, the dissolution free energy was evaluated as

$$\begin{aligned}\Delta G_{\mathrm{diss}}(U) = {} & E_{\mathrm{vac}} - E_{\mathrm{intact}} + \mu_{\mathrm{Ir,bulk}} + 3e(E^{\circ}_{\mathrm{Ir^{3+}/Ir}} - U) \\ & + m[\mu_{\mathrm{O}}(U) - \delta G_{\mathrm{O}}] + \Delta\Delta G_{\mathrm{solv}},\end{aligned} \tag{9}$$

where $\mu_{\mathrm{O}}(U) = G_{\mathrm{H_2O}} - G_{\mathrm{H_2}} + 2eU$, $E^{\circ}_{\mathrm{Ir^{3+}/Ir}} = 1.156$ V,[46] and $\mu_{\mathrm{Ir,bulk}}$ was obtained from a same-method 108-atom bulk-Ir calculation. The correction for each co-removed Ir-exclusive O was $\delta G_{\mathrm{O}} = 0.05$ eV. Solvation was included through the difference between the SCCS corrections of the vacancy and intact endpoints,

$$\begin{aligned}\Delta\Delta G_{\mathrm{solv}} = {} & [E_{\mathrm{SCCS}}(\mathrm{vac}) - E_{\mathrm{vacuum}}(\mathrm{vac})] \\ & - [E_{\mathrm{SCCS}}(\mathrm{intact}) - E_{\mathrm{vacuum}}(\mathrm{intact})].\end{aligned} \tag{10}$$

Positive ΔG_{diss} indicates that the intact structure is thermodynamically favored relative to the corresponding vacancy and Ir^{3+}(aq) products. At pH 0, the numerical SHE and RHE scales coincide, and potentials are reported versus RHE for consistency with the OER analysis. Because the Ir^{3+}/Ir reference contributes a $-3e$ potential dependence while each co-removed O contributes a $+2e$ CHE term, the potential dependence of each local dissolution process is

$$\frac{d\Delta G_{\mathrm{diss}}}{dU} = (2m - 3)e. \tag{11}$$

The primary analysis included 413 sites with converged SCCS dissolution energetics and chemically retained vacancy structures. Local structural descriptors were evaluated on the corresponding intact structures. To characterize the oxygen coordination of the dissolving Ir center, we defined the distance-weighted quantity

$$S_{\mathrm{O}} = n_{\mathrm{O}} \exp\left[\frac{1.9\ \text{Å} - \langle r_{\mathrm{Ir-O}} \rangle}{0.37\ \text{Å}}\right], \tag{12}$$

which incorporates both the number and mean distance of neighboring O atoms. Correlations were evaluated both across all sites and within individual parent structures. For comparison with OER activity, sites were matched to the corresponding *O pathways; when multiple *O centers shared the same host Ir, the pathway with the lowest SCCS overpotential was used. Additional details of the vacancy construction, quality-control criteria, potential dependence, and activity–stability analysis are provided in Supporting Information, Section S8.

4.4 Structural descriptors and electronic-structure analysis

Structural descriptors were evaluated on the relaxed working *O structures and were selected to represent the local coordination of the active Ir–O unit, Ir connectivity, interaction with the support, and local oxygen environment. The primary descriptor, $\mathrm{CN}_{\mathrm{O,co}}(\mathrm{Ir}^{*})$, was defined as the number of O atoms other than O* satisfying both Ir*–O < 2.5 Å and O*–O < 2.9 Å. It therefore counts O co-ligands of the binding Ir* that also lie within the immediate geometric environment of O*. The distribution of $\mathrm{CN}_{\mathrm{O,co}}(\mathrm{Ir}^{*})$=0–4 was evaluated over all 303 complete pathways for each cluster size.

Eight nonconstant structural quantities were included in the descriptor comparison: host-Ir oxygen coordination, direct Ir–Ir coordination, Ir network degree, Ir–Ti contacts, cluster-relative Ir height, $\mathrm{CN}_{\mathrm{O,co}}(\mathrm{Ir}^{*})$, mean Ir–O distance, and local O/Ir ratio. Supporting Information, Section S5 and Tables S4–S7 define the descriptor library and report its variability, grouped validation, and feature-selection stability; Section S6 and Table S8 provide the size-resolved census of all 465 Ir microenvironments.

Pearson and Spearman correlations were evaluated across all sites and after subtracting the corresponding size-dependent mean from each descriptor and target quantity. Uncertainty in the primary Spearman correlations was estimated using 2000 clustered bootstrap replicates in which the 30 Ir cluster sizes were resampled with replacement.

Predictive transfer across cluster sizes was evaluated using leave-one-size-out cross-validation. In each outer fold, all sites belonging to one Ir cluster size were excluded from model fitting and used for testing. Ridge and elastic-net models were optimized by grouped cross-validation within the remaining cluster sizes. Feature standardization and model selection were performed using only the training data in each fold. No random site-level splitting was used because sites belonging to the same working structure are not statistically independent. Model hyperparameters and additional statistical analyses are provided in the Supporting Information.

Electronic properties were evaluated by single-point CP2K calculations on the SCCS working *O structures using the same geometries and electronic structure settings as the thermochemical calculations. Hirshfeld populations were evaluated for the active O* and binding Ir* atoms across all 303 complete pathways.

Mayer bond orders were evaluated for representative active Ir–O pairs on Ir_{15} and Ir_{17}. Converged SCCS orbitals were exported from CP2K in Molden format and analyzed using Multiwfn 3.8dev.[47, 48] For representative Ir_{17} sites with $\mathrm{CN}_{\mathrm{O,co}}(\mathrm{Ir}^{*})$ = 0 and 3, atom- and angular-momentum-resolved densities of states were evaluated for the active O and binding Ir atoms. The O 2p and Ir 5d contributions were referenced to the calculated Fermi level and broadened using a Gaussian width of 0.12 eV. No additional geometry relaxation was performed for these electronic- structure analyses.

4.5 Steady-State Microkinetic Model

The four AEM adsorbate states were treated as a reversible kinetic cycle. For an elementary step with reaction free energy $\Delta G(U)$, the forward activation barrier was described using the three-regime electrochemical

barrier construction of Mao *et al.*[38]:

$$E_a = \begin{cases} 0, & \Delta G < -E_a^0/\beta, \\ E_a^0 + \beta\Delta G, & -E_a^0/\beta \le \Delta G \le E_a^0/(1-\beta), \\ \Delta G, & \Delta G > E_a^0/(1-\beta). \end{cases} \tag{13}$$

The forward rate constant was calculated as

$$k_f = A \exp(-E_a/k_\mathrm{B}T), \tag{14}$$

and the reverse rate was determined by detailed balance. The baseline parameters were $A = 10^9$ s^{-1}, $E_a^0 = 0.26$ eV, $\beta = 0.5$, and $T = 298.15$ K. Steady-state intermediate populations were obtained by solving the four-state linear master equation, and the net cycle rate was evaluated from the forward-minus-reverse flux between adjacent states.

Polarization curves were calculated from 1.00 to 2.00 V using a 0.004 V potential interval. Current densities were obtained from

$$j = 4\rho_e \mathrm{TOF}, \tag{15}$$

using $\rho_e = 78.3$ μC cm^{-2} as the reference site-density factor. Kinetic overpotentials were interpolated at 10 mA cm^{-2}, and apparent Tafel slopes were fitted over 0.1–10 mA cm^{-2}. For each cluster size, the ensemble rate was obtained by summing the rates of all complete anodic sites. Sensitivity to the assumed electrochemical barrier parameters was evaluated over $E_a^0 = 0.15$–0.80 eV and $\beta = 0.4$–0.6; Supporting Information, Section S7, Figure S6, and Table S10 report the resulting rank retention, target-reaching site counts, and best-size/onset changes.

Acknowledgements

W.-L.L. acknowledges support from the National Science Foundation under Award No. CBET-2534493 and the ACS Petroleum Research Fund under Doctoral New Investigator Grant 69037-DNI10. W.-L.L. also acknowledges the Hellman Fellowship at the University of California San Diego. The calculations were performed primarily on Expanse at the San Diego Supercomputer Center (SDSC) through the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program. Additional calculations were performed by A.A. using computational resources provided by the KAUST Supercomputing Core Laboratory under Project K10175.

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

G.L. and L.Z. contributed equally to this work. G.L. and L.Z. performed the calculations, analyzed the data, prepared the figures and wrote the manuscript. X.M. analyzed the data and prepared the figures. A.A. contributed to the calculations. H.Z. contributed to the interpretation of the results and discussions. W.-L.L. conceived and supervised the project and directed the research. All authors discussed the results and reviewed the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the corresponding author.

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